Session 6: Knowledge Representation & Reasoning

Last session:

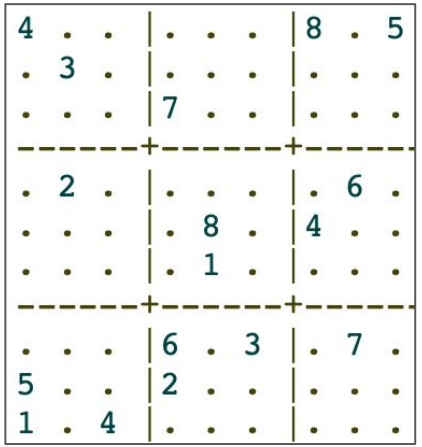
- The “magic” of constraint propagation

This session:

- Augmenting a constraint network with knowledge representations (a “**knowledge base**”), and augmenting basic propagation mechanisms with reasoning strategies (an “**inference engine**”).

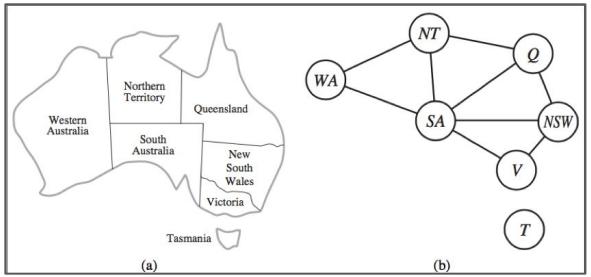
Key idea: intelligence requires first getting to know what the problem is. Hence, assigning credit = an epistemological process

Heuristic approach: problem is given: we know how to propagate, and if we get stuck, we can always just guess



Epistemological approach: imagine we don't even know this is Sudoku. How do we find out?

John McCarthy: we get to know problems based on our knowledge of stable phenomena and ability to reason about specific situations



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- **Phenomena**: e.g., knowledge of games, propagation, in general, etc.

- **Situations**: e.g., reasoning about what the constraints are on a particular Sudoku game.

McCarthy’s core idea

- Argued intelligence depends on representing **commonsense knowledge about the world** and reasoning about this knowledge.
- This knowledge should not be tied to a single problem.
- Knowledge should be reusable across many different situations.

John McCarthy (1927-2011), co-founder of MIT AI lab and founder of Stanford AI lab

Source: https://aitoolsexplorer.com/ai-history/john-mccarthy-the-father-of-artificial-intelligence/

Search trees, constraint networks etc. are not stable, since situations change. Phenomena are stable.

Situations (states of the “universe”)

- A specific context in which problems arise and actions occur.
- Situations include current facts, goals, and environment.

Phenomena (stable facts and concepts)

- General, stable facts about how the world works.
- Examples include physical and everyday regularities.

*Learning to assign credit = learning to **represent** and **reason** about **knowledge** of **phenomena** regarding changing, real-world **situations**.*

Supermarket example: how to send targeted ads to a shopper who visits rarely, but always buys frozen spinach?

- Standard way: use data science to analyze data in receipts + customer info.
- McCarthy's way: use commonsense knowledge to interpret behavior:
  - The shopper likely has a freezer.
  - The shopper likes the frozen spinach more than the place she/he usually shops at.

Supermarket example: McCarthy argues KRR = greater **robustness**, **efficiency**

- **Robustness** to changes in customer's "situation":
  - How much space is currently in the shopper's freezer, whether relatives are visiting, etc.
- **Efficiency:**
  - Standard data science: only works well with vast volumes of data + big computers to analyze data (brute force)
  - McCarthy's way (KRR): just need a representation of commonsense knowledge + a few facts about a customer.

How could an AI system or org. reason about the vast number of possible situations?

- McCarthy's answer: this is possible if we can represent **commonsense knowledge** about phenomena and situations, and some way of **reasoning** about this knowledge.

- Before we talk about what these mean, let us talk about the nature of credit assignment in KRR.

Credit assignment & knowledge representation and reasoning

(1) The vision: an AI system is intelligent when it behaves like an “**Advice taker**”
(2) The credit assignment challenge of the “Advice taker” = the “**frame problem**”

McCarthy's "Advice taker" vision (1959)

Advice taker = a hypothetical AI system. Think of an LLM… but (in McCarthy's view at least), something much better. Some criteria:

- (1) It learns new representations of problem just by you telling it **facts** (e.g., not prompts!) about the situation in natural language (English, Korean, etc.) = it takes advice
- (2) It represents knowledge as **concepts** and **facts**, not procedures, algorithms.
- (3) It precisely and logically reasons about these concepts and facts to modify its representations (e.g., no hallucination, no slop, no generic outputs).

McCarthy’s “Advice taker” - Requirements

- Ability to represent commonsense knowledge about any relevant phenomena
- Ability to reason about this knowledge across situations

The frame problem

What are the challenges of achieving McCarthy's Advice taker vision?

- Obvious one: there is a lot of commonsense knowledge about the world

McCarthy: actually the challenge is much more severe

- We need knowledge not just of what to do and when to do it
- We need knowledge of what not to do = the frame problem

Illustrating the frame problem - what did Ahn Junghwan need to “know” to score a goal against Italy? (Daejeon, World Cup 2002)

Source: “**My favourite game: when Ahn Jung-hwan eliminated Italy from the World Cup”,** https://www.theguardian.com/football/2020/apr/26/my-favourite-game-when-ahn-jung-hwan-eliminated-italy-from-world-cup

Solving the frame problem

Requires:

- Ability to represent commonsense knowledge about any relevant phenomena
- Ability to reason about this knowledge across situations

Representing commonsense knowledge: how can we do this efficiently given how much of it there is?

- To see how, let us distinguish explicit vs. implicit knowledge

Consider the following sentence: why do you believe it?

“The Pyramids of Giza in Egypt are older than my neighbor’s house”

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“The Pyramids of Giza in Egypt are older than my neighbor’s house”

→ Answer: it was **logically entailed**, meaning that it was implicit in your commonsense knowledge even though you never explicitly thought about it before.

Implicit knowledge: implications for representing commonsense knowledge

- (1) Knowledge representations should be **declarative**
- (2) Knowledge representations need to allow **concepts**
- (3) Knowledge representations need to reflect **contexts**

(1) Knowledge representations need to be declarative

- What was implicit was based not on sequences (e.g., sequences of words in that sentence), but **facts** and **concepts**.

- Hence, commonsense knowledge should be represented **declaratively** (e.g., as **lists**) not procedurally (e.g., as algorithms — most programming languages).

Declarative representations - omelette example: can more easily change the “recipe” on the right than the one on the left

Procedural representation: step-by-step sequence — changing one step may change the whole recipe.

Declarative representation: changing one “fact” may still enable changing the overall recipe fairly simply.

(2) Knowledge representations need to allow concepts

- **Concepts**. Not just facts or programs: ways of talking about problems at different **levels of abstraction**.

- Our implicit knowledge told us that the Pyramids of Giza should be older than my neighbor's house since we have abstract concepts of pyramids and of a neighbor's house.

(3) Knowledge representations need to reflect contexts

- **Contexts**. the meaning of our knowledge representations depends on our situations and goals — the context.

- Consider the following sentence:

  "*I tried to fill the suitcase with the whale, but it was too big*".

- What is "it"? How do you know?

(3) Knowledge representations need to reflect contexts

- **Speech acts**: knowledge representations can be used to indicate actions to be performed. In this case, the meaning of the knowledge representation depends on who is using it (their goals, etc.).

- Example:
  - Consider the phrase, “Apple (the company) needs to perform better”.

Elaboration: adding or removing facts, concepts etc. from a knowledge base

- Elaboration involves “**operations**” on a knowledge base
  - Syntactic operations: putting knowledge in the correct format.
  - Semantic operations: modifying which concepts or facts are valid or relevant.
  - These operations are costly.
  - “Garbage” knowledge representations can easily pile up in a knowledge base.
  - McCarthy: intelligence requires some way to efficiently “take out the trash” = “**elaboration tolerance**”

Elaboration tolerance

The effort required to add new [facts and concepts] to the [knowledge base] is proportional to the complexity of the [facts and concepts].

Low elaboration tolerance is akin to “digital clutter”

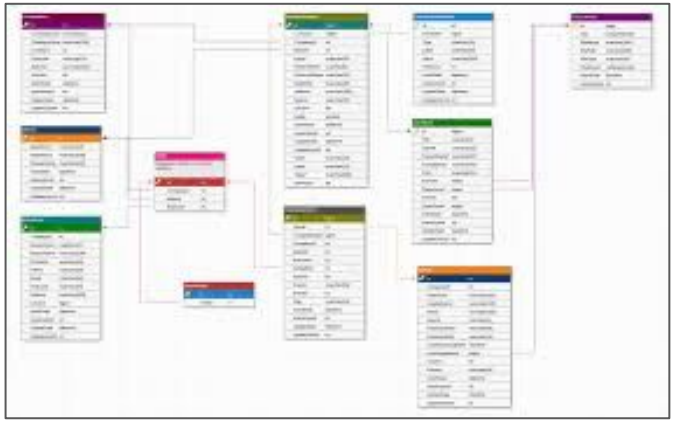
Source: https://pepworldwide.biz/blog/how-digital-clutter-affects-you-and-what-to-do-about-it/

Elaboration tolerance requires **default reasoning** = making commonsense assumptions, even if we are not sure

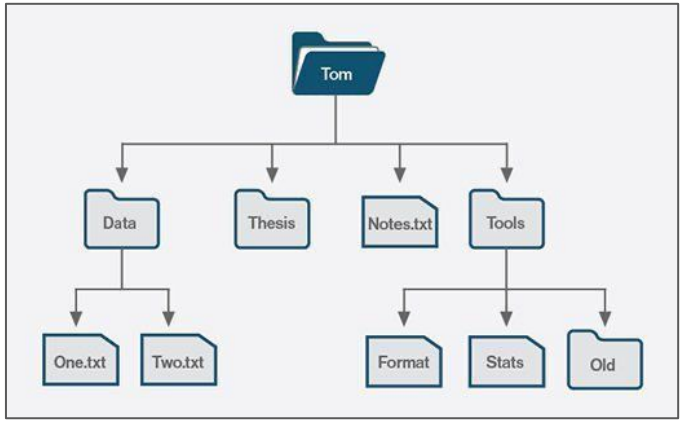
Example: “*I tried to fill the suitcase with the whale, but it was too big*”

→ It *could* be a stuffed animal, or a baby whale, or a lunchbox with a picture of whale on it… but we need to make some assumption to simplify things.

Elaboration tolerance requires adequately **expressive languages**.
Example: SQL database versus file system.



SQL = express logical relations to enable high data integrity



File system = lower expressiveness, and lower data integrity.

Approaches to knowledge representation and reasoning

(1) Deep learning approach

(2) Mathematical logic approach

(1) Deep learning approach to KRR

- Knowledge representation:
  - Bit string patterns extracted from large volumes of data and stored in matrix-like structures (vector databases, knowledge graphs, etc.)
  - Concepts = statistical relations among the bit strings
  - Context = adding files using techniques such as retrieval-augmented generation (RAG) and context engineering.
- Reasoning: heuristics for prompting
  - E.g.: chain-of-thought procedures

(1) Deep learning approach to KRR (continued)

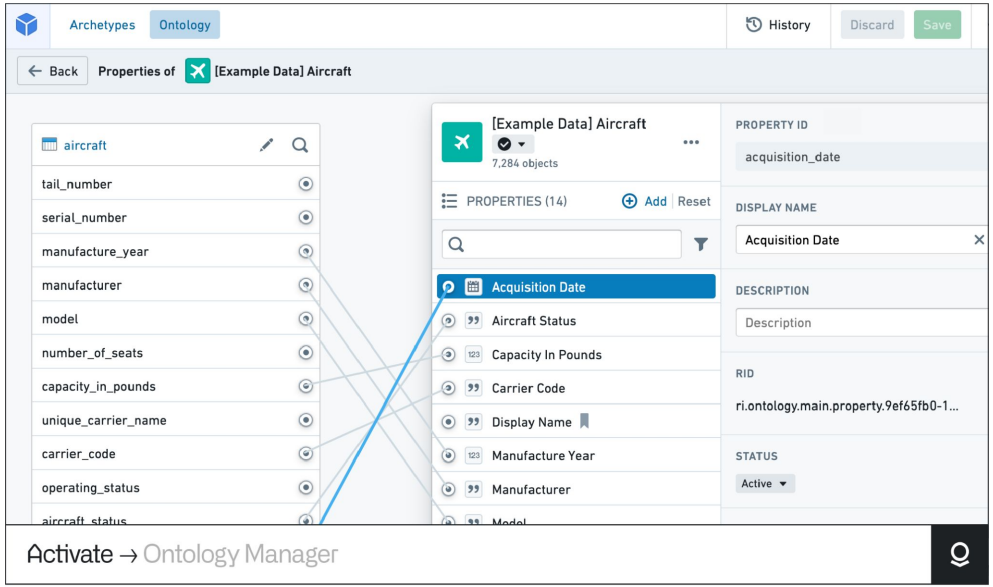
- Reasoning: heuristics for prompting
  - E.g.: chain-of-thought procedures
- Limits:
  - So far: little support for solving the frame problem or elaboration tolerance

(2) Mathematical logic approach (McCarthy's approach)

- Mathematical logic = precise, formal languages for describing facts and concepts etc. and how to reason about them.
  - Often caricatured as just "if-then rules".

- Knowledge is represented using rich conceptual structures (e.g., **ontologies**).
  - The goal is to make knowledge transparent to reason step-by-step, not heuristically through prompting.

(2) Mathematical logic approach (Palantir example)



Source: https://www.palantir.com/explore/platforms/foundry/ontology/